Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

Constructor Group *Machine Learning Consultant* **2021–present**

National University of Singapore *Postdoc under Kostya Novoselov* **2022–present**

Wyckoff Transformer, a generative model for materials ICML 2025 · GitHub

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Cut inference to 0.05 GPU-ms/structure — 4 orders of magnitude faster than prior models
- Implemented the model in PyTorch; production-ready package published on PyPI
- Orchestrated large-scale (10k structures) DFT computations with VASP on an HPC cluster
- Led a team of 6

ML for defects in 2D crystals npj 2D Mater. Appl. · npj Comput. Mater. · patent · GitHub

- Conceived a novel sparse representation of crystals with defects
- Cut energy-prediction error 3.7× vs. the best prior model
- Optimised training and inference: 4x less memory, 8x fewer GPU operations
- Built a reproducible pipeline for parallel ML experiments
- Led a team of 3

⟨CERN | Yandex → HSE University⟩ *Researcher* **2014–2022**

2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.

Generative models for fast simulation paper 1 · paper 2 · paper 3 · code

- Built a GAN for high-fidelity, tabular-data simulation of a Cherenkov detector
- Reached $10^5\times$ speed-up over the full Geant4 simulation

Machine learning on noisy data paper 1 · paper 2 · code

- Derived a statistically rigorous method to train classifiers on sPlot-weighted (label-noisy) data, standard in high-energy physics

Muon identification at the LHCb experiment at CERN paper · poster · IDAO-2019

- Solved a classification problem over tabular data with noisy labels under timing constraints
- Built a CatBoost model that cut the false-positive rate 30% in the critical low-momentum region
- Shipped the model into the LHCb trigger (C++ & Python)
- Packaged the work as a data science competition problem for IDAO-2019

Education

HSE University & Sapienza Università di Roma **2016–2020**
PhD in Computer Science and Physics (Double Degree)
Supervisors: Andrey Ustyuzhanin & Barbara Sciascia
Thesis: Machine Learning for particle identification in the LHCb detector

Product Management course at Yandex **2018**
Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**
Master's-level CS programme
Algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics **2014–2016**
Optimisation of data processing of the LHCb experiment.
BS in Physics **2010–2014**
Wave-packet molecular dynamics of electrons in nonideal plasma.

International Junior Science Olympiad (IJSO) **Korea, 2008**
Silver medal

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 2).

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a science slam, interviews (1, 2), and a public talk.
- Taught at the Machine Learning in High Energy Physics Summer School in 2015, 2016, 2017, 2018, 2019, 2020, 2021, and 2023.
- Lecturer in the award-winning Coursera Advanced Machine Learning specialisation
- Delivered more than 20 conference talks.

Service

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

- **Generative modelling for scientific data** — GANs, autoregressive transformers, and diffusion; from fast detector simulation to symmetry-aware crystal generation.
- **Uncertainty quantification** — built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **ML on structured, non-text data** — tabular, graph, and atomistic representations, spanning gradient boosting to transformers.
- **Inference and systems at scale** — C++ production integration, large-scale DFT/VASP on HPC clusters, and reproducible experiment pipelines.
- **Physics domain depth** — high-energy and condensed-matter physics; first-principles intuition that shapes model design.
- **Research leadership** — lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • C++ • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

Presenting at ICML 2026

July 11, 12:15–13:30 — **Nikita Kazeev** and Ian Babich. "Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments". In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 10:50–11:50 & 15:55–16:55 — Artem Dembitskiy*, Shuya Yamazaki*, Artem Maevskiy*, **Nikita Kazeev***, Roman A. Eremin, Semen Budennyy, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. "Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials". In: *AI for Science Workshop at ICML 2026*. ***Equal Contribution**.

Selected Publications

Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hippalgaonkar. "Wyckoff Transformer: Generation of Symmetric Crystals". In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Nikita Kazeev, A. R. Al-Maeeni, I. Romanov, et al. "Sparse representation for machine learning the properties of defects in 2D materials". In: *npj Computational Materials* 9 (2023), p. 113.

Denis Derkach, **Nikita Kazeev**, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. "Cherenkov detectors fast simulation using neural networks". In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). **Alphabetic order, I'm the corresponding author.**